Questions: Introduction to t-testing

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Summary

A selection of questions for the study guide on introduction to t-testing.

Before attempting these questions, it is highly recommended that you read Guide: Introduction to t-testing.

Q1

1.1. What is the *rule of thumb* for sample size in t-testing?

1.2. What is the definition of a t-test?

1.3. What is the definition of a Student's t-test?

1.4. What is the definition of a Welch's t-test?

Q2

Fill in the tables below and identify what each symbol stands for in these test statistics.

2.1. A one-sample test statistic

Symbol	Meaning
$\bar{x}$	
μ_0	
s	
n	

2.2. A two-sample test statistic

Symbol	Meaning
μ_1	

Symbol	Meaning
μ_2	
s_1	
s_2	
n_1	
n_2	

2.3. What is the relationship between s_1 and s_2 for:

a) A Student's t-test?

b) A Welch's t-test?

Q3

The following examples will follow sweet shops named Cantor's Confectionery and Lovelace's Lollies. For the examples, construct a null and alternative hypothesis.

3.1. A one-sample, one-tailed t-test where the sample mean is $\bar{x}$. Cantor's Confectionery want to test if their mean daily earnings are greater than the market average μ_0.

3.2. A two-sample, one-tailed t-test where Cantor's Confectionery compare the mean daily earnings, (μ_1) to Lovelace's Lollies' (μ_2). They want to test if they earns more than Lovelace's Lollies do.

3.3. A two-sample, two-tailed t-test where Cantor's Confectionery compare the mean earnings from weekday sales (μ_1) and weekend sales (μ_2) to test if there is a difference.

Q4

4.1. Use the information below to calculate a test statistic.

Symbol	Value
$\bar{x}$	10
μ_0	7
s	0.5
n	19

4.1.1 Is this a Students t test or a Welch's t test?

4.1.2 Why?

4.2. Use the information below to calculate a test statistic.

Symbol	Value
μ_1	12
μ_2	13
$s_1 = s_2$	2
n_1	15
n_2	16

4.2.1 Is this a Students t test or a Welch's t test?

4.2.2 Why?

4.3. Use the information below to calculate a test statistic.

Symbol	Value
μ_1	45
μ_2	51
s_1	3
s_2	3.5
n_1	19
n_2	18

4.3.1 Is this a Students t test or a Welch's t test?

4.3.2 Why?

Q5

5.1. Perform a t-test on the following scenario:

Cantor's Confectionery want to perform a t-test at the 5% significance level. They want to test if their average daily earnings are more than the average for sweet shops in the area. They record the earnings for one week (7 days) and calculate the sample mean daily earnings as £905, with sample standard deviation £4. The market average is £901.20.

5.1. Perform a t-test on the following scenario:

Cantor's Confectionery want to test whether weekday sales and weekend sales are different. They record daily earnings over several weeks and calculate the following statistics. The test is conducted at the 5% significance level.

5.2. Cantor's Confectionery introduce a new shop window display and want to test if it increases average daily earnings compared to the old display. They collect data on daily earnings before and after the change. The test is conducted at the 5% significance level.

After attempting the questions above, please click this link to find the answers.

Version history and licensing

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